Sensors & Navigation On-boarding Tasks

General Prerequisites (Applies for all sub-teams):

- Completed Setting Up Your Development Environment
- Have a working knowledge of C++ and ROS2

Instructions: Students wishing to be on-boarded to the team should solve 1 problem. This is NOT a meaningless interview-style task. Instead, these problems are actual problems the sub-team is struggling with. This is your first real task on the sub-team. Students should refrain from completing multiple problems without prior approval to ensure all students wishing to be on-boarded have a task to complete. **Upon completion of problem, please meet with sub-team lead prior to pushing code changes.**

Increasing the Reliability & Accuracy of Sensor Nodes

Problem 1: From logging arbitrary status numbers to failing to exit cleanly, the reliability of our sensor nodes needs to be increased in order to be ready to support autonomous navigation tasks.

- Languages Required: C++, minimal C
- Frameworks Required: ROS2
- Suggested Approaches: creating logical and well-documented statuses, testing edge cases to ensure Ctrl+C always ends execution

Problem 2: Particularly our IMU has been identified as inconsistent and in-accurate (including inconsistently re-zeroing itself). These inconsistencies must be identified, traced, and fixed if possible and clearly documented if not.

- Languages Required: C++, minimal C
- Frameworks Required: ROS2
- Suggested Approaches: first performing physical tests of the IMU and systematically eliminating possible ways that error and inconsistencies are being introduced

Navigation

Problem 1:

- Languages Required:
- Frameworks Required:
- Suggested Approaches:

Problem 2:

- Languages Required:
- Frameworks Required:
- Suggested Approaches: